CIVIL ENGINEERING / PORTFOLIO / 2026

MD

Miechilly Duong



CIVIL ENGINEERING STUDENT

B.S. Civil Engineering | Geotechnical Concentration

Kennesaw State University • GPA 3.6 • May 2029

GEOTECHNICAL / INFRASTRUCTURE / CONSTRUCTION

OPEN TO SUMMER 2027 INTERNSHIPS & CO-OPS

About & Portfolio Guide

Civil engineering student with experience in undergraduate research, engineering software, structural model fabrication, and hands-on welding. I am interested in how materials, soil behavior, design, and construction methods come together in real infrastructure projects.

EDUCATION

Kennesaw State University

B.S. Civil Engineering — Geotechnical Concentration

3.6 GPA • Expected May 2029

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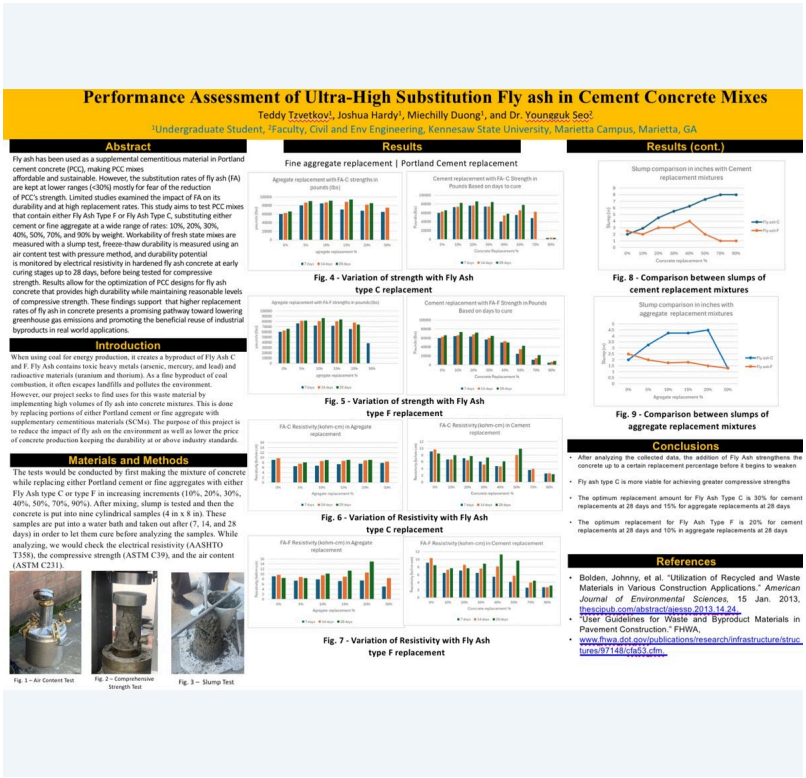
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Technical Skills

AutoCAD • Civil 3D • MATLAB • Excel • Arduino • Research data analysis • Technical writing • Welding and fabrication • English and Vietnamese

High-Volume Fly Ash Concrete

Studying the effect of Class C and Class F fly ash replacement on concrete properties, with an emphasis on sustainable materials and practical performance.



Performance Assessment of Ultra-High Substitution Fly ash in Cement Concrete Mixes

Teddy Tzvetkov[1], Joshua Hardy[1], Miechilly Duong[1], and Dr. Younguk Seo[2]

[1]Undergraduate Student, [2]Faculty, Civil and Env Engineering, Kennesaw State University, Marietta Campus, Marietta, GA

Abstract

Fly ash has been used as a supplemental cementitious material in Portland cement concrete (PCC), making PCC mixes affordable and sustainable. However, the substitution rates of fly ash (FA) are kept at lower ranges (<30%) mostly for fear of the reduction of PCC's strength. Limited studies examined the impact of FA on its durability and at high replacement rates. This study aims to test PCC mixes that contain either Fly Ash Type F or Fly Ash Type C, substituting either cement or fine aggregate at a wide range of rates: 10%, 20%, 30%, 40%, 50%, 70%, and 90% by weight. Workability of fresh state mixes are measured with a slump test, freeze-thaw durability is measured using an air content test with pressure method, and durability potential is monitored by electrical resistivity in hardened fly ash concrete at early curing stages up to 28 days, before being tested for compressive strength. Results allow for the optimization of PCC designs for fly ash concrete that provides high durability while maintaining reasonable levels of compressive strength. These findings support that higher replacement rates of fly ash in concrete presents a promising pathway toward lowering greenhouse gas emissions and promoting the beneficial reuse of industrial byproducts in real world applications.

Introduction

When using coal for energy production, it creates a byproduct of Fly Ash C and F. Fly Ash contains toxic heavy metals (arsenic, mercury, and lead) and radioactive materials (uranium and thorium). As a fine byproduct of coal combustion, it often escapes landfills and pollutes the environment. However, our project seeks to find uses for this waste material by implementing high volumes of fly ash into concrete mixtures. This is done by replacing portions of either Portland cement or fine aggregate with supplementary cementitious materials (SCMs). The purpose of this project is to reduce the impact of fly ash on the environment as well as lower the price of concrete production keeping the durability at or above industry standards.

Materials and Methods

The tests would be conducted by first making the mixture of concrete while replacing either Portland cement or fine aggregates with either Fly Ash type C or type F in increasing increments (10%, 20%, 30%, 40%, 50%, 70%, 90%). After mixing, slump is tested and then the concrete is put into nine cylindrical samples (4 in x 8 in). These samples are put into a water bath and taken out after (7, 14, and 28 days) in order to let them cure before analyzing the samples. While analyzing, we would check the electrical resistivity (AASHTO T358), the compressive strength (ASTM C39), and the air content (ASTM C231).

Fig. 1 – Air Content Test

Fig. 2 – Comprehensive Strength Test

Fig. 3 – Slump Test

Results

Fine aggregate replacement | Portland Cement replacement

Fig. 4 - Variation of strength with Fly Ash type C replacement

Fig. 5 - Variation of strength with Fly Ash type F replacement

Fig. 6 - Variation of Resistivity with Fly Ash type C replacement

Fig. 7 - Variation of Resistivity with Fly Ash type F replacement

Results (cont.)

Fig. 8 - Comparison between slumps of cement replacement mixtures

Fig. 9 - Comparison between slumps of aggregate replacement mixtures

Conclusions

- After analyzing the collected data, the addition of Fly Ash strengthens the concrete up to a certain replacement percentage before it begins to weaken
- Fly ash type C is more viable for achieving greater compressive strengths
- The optimum replacement amount for Fly Ash Type C is 30% for cement replacements at 28 days and 15% for aggregate replacements at 28 days
- The optimum replacement for Fly Ash Type F is 20% for cement replacements at 28 days and 10% in aggregate replacements at 28 days

References

- Bolden, Johnny, et al. "Utilization of Recycled and Waste Materials in Various Construction Applications." *American Journal of Environmental Sciences*, 15 Jan. 2013, thescipub.com/abstract/ajessp.2013.14.24.
- "User Guidelines for Waste and Byproduct Materials in Pavement Construction." FHWA, www.fhwa.dot.gov/publications/research/infrastructure/structures/97148/cfa53.cfm.





Research approach

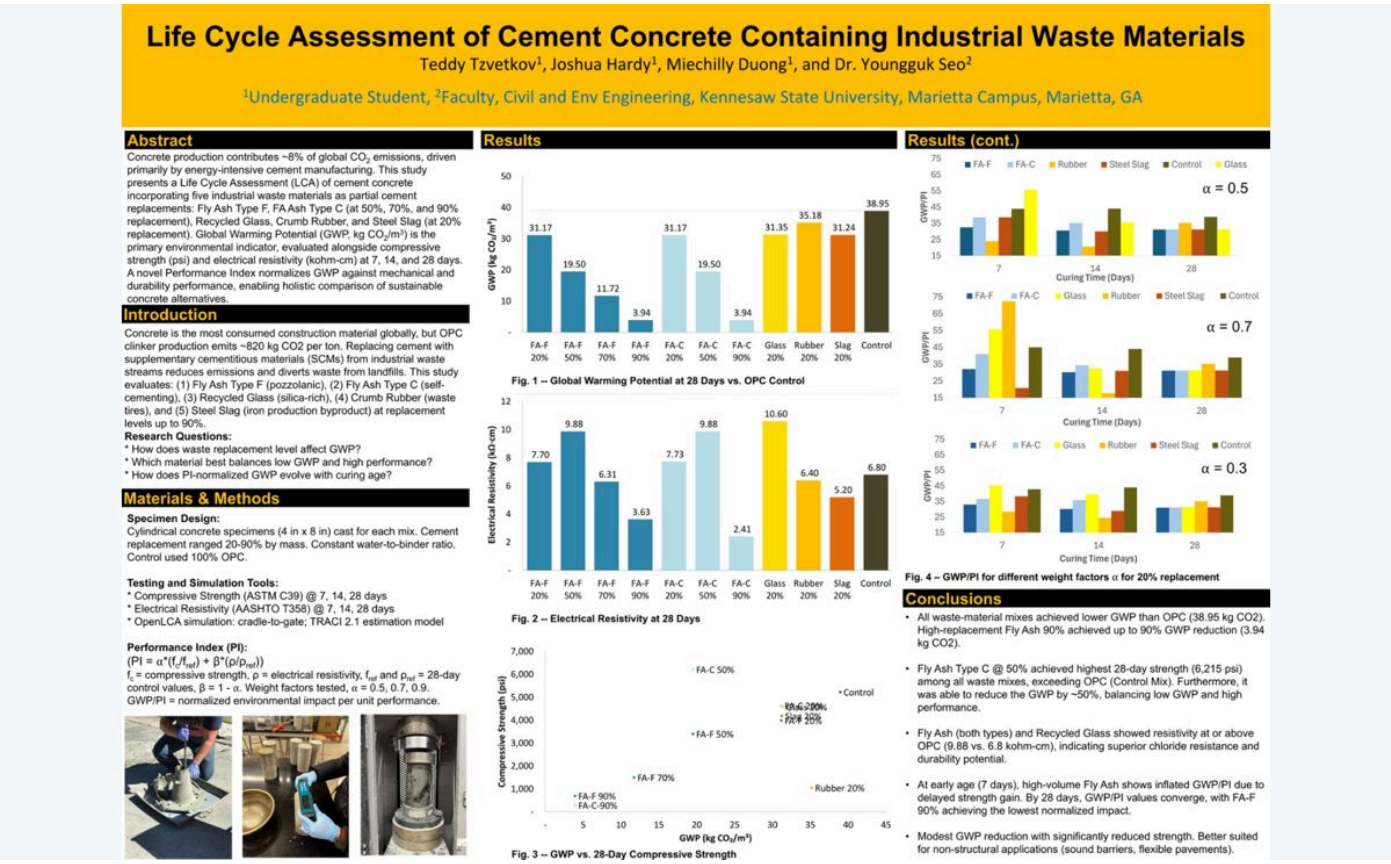
Prepared and evaluated concrete mixtures with varying fly ash substitution levels. The broader study considered workability, compressive strength, and electrical resistivity as indicators of performance.

Engineering takeaway

Lower cement content can reduce environmental impacts, but material selection must also account for strength, durability, and construction practicality. Numerical findings and testing conditions are documented in the original research poster.

Life-Cycle Assessment

Comparing the environmental footprint of cement-based materials with industrial byproducts while considering the engineering performance of the resulting concrete.



Materials evaluated

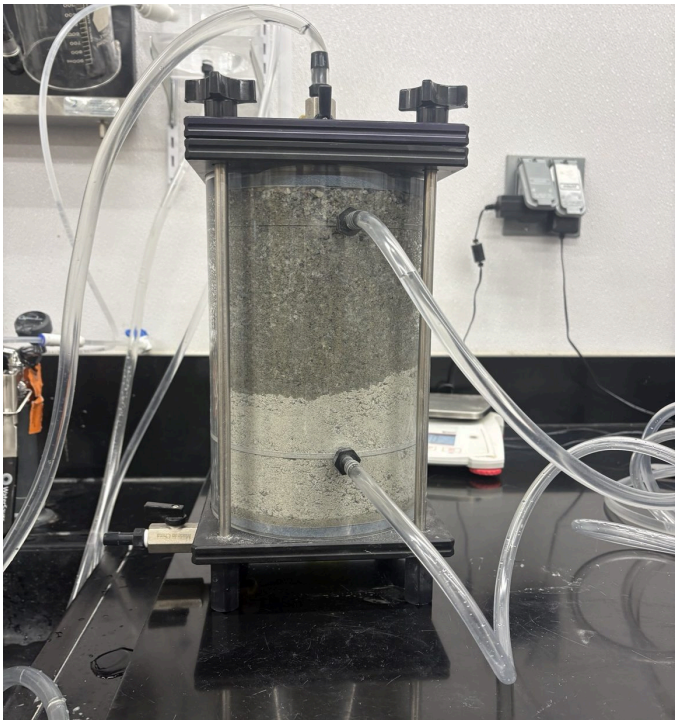
The research considers alternative constituents such as fly ash, recycled glass, crumb rubber, and steel slag in the broader context of sustainable concrete.

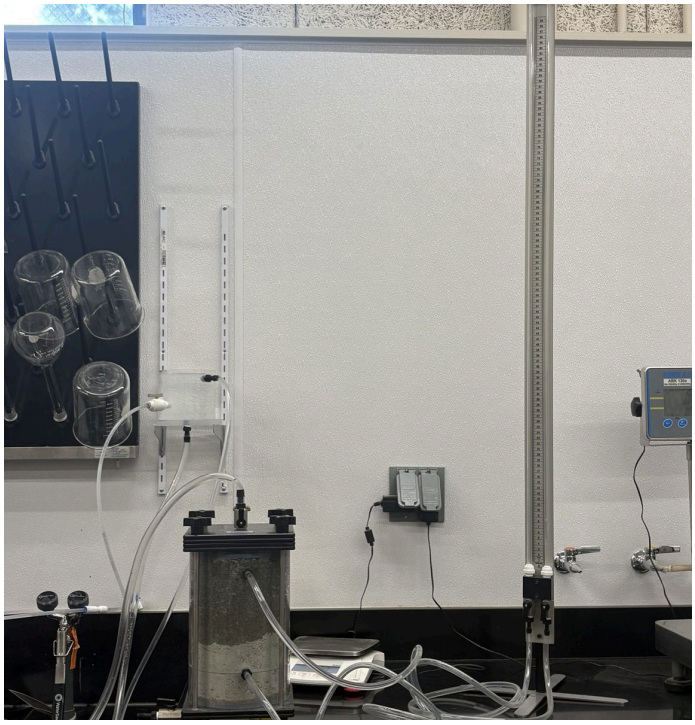
Why it matters

Environmental performance alone is not enough to judge whether a concrete mixture is useful. A credible comparison weighs material impacts alongside mechanical properties and suitability for infrastructure.

MSE Wall Backfill Soil

Laboratory characterization of granular backfill materials for mechanically stabilized earth (MSE) walls, emphasizing particle-size distribution and hydraulic conductivity.





Laboratory work

Prepared soil specimens and examined gradation and permeability behavior to understand how backfill composition influences drainage and potential suitability for retaining-wall applications.

Interpreting results

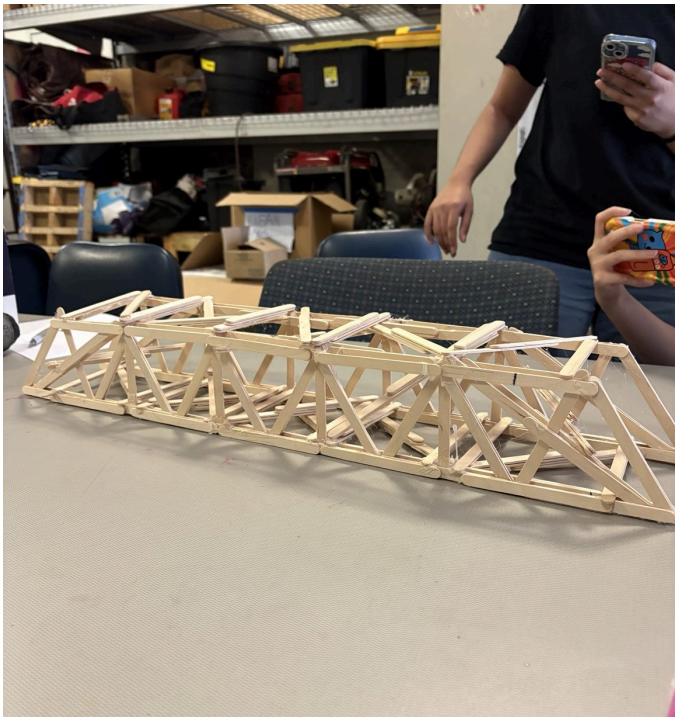
Gradation and fines content can affect water movement through backfill. Results remain part of ongoing characterization; laboratory measurements should be verified before drawing design conclusions.

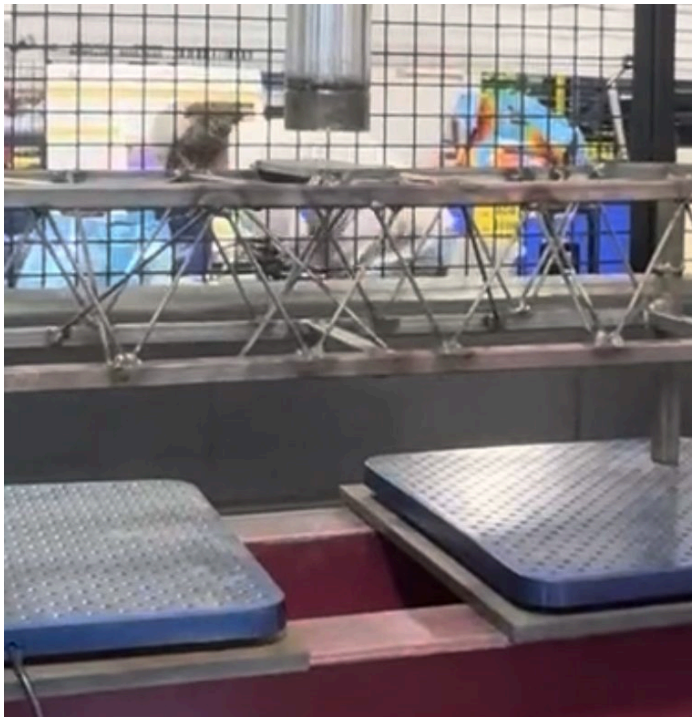
Engineering relevance

Drainage performance is an important consideration in evaluating granular materials for reinforced-soil infrastructure.

Steel Bridge Team

An ongoing team project translating a popsicle-stick truss bridge model into a steel replica to study structural arrangement, connections, and fabrication.





From concept to steel

Our team first assembled a lightweight physical model with triangular bracing, then recreated its overall configuration with steel members. The two versions allow us to compare member layout and practical assembly considerations.

What I am learning

The work provides experience with structural visualization, alignment at joints, collaborative problem-solving, and the differences between a small-scale concept and metal fabrication.

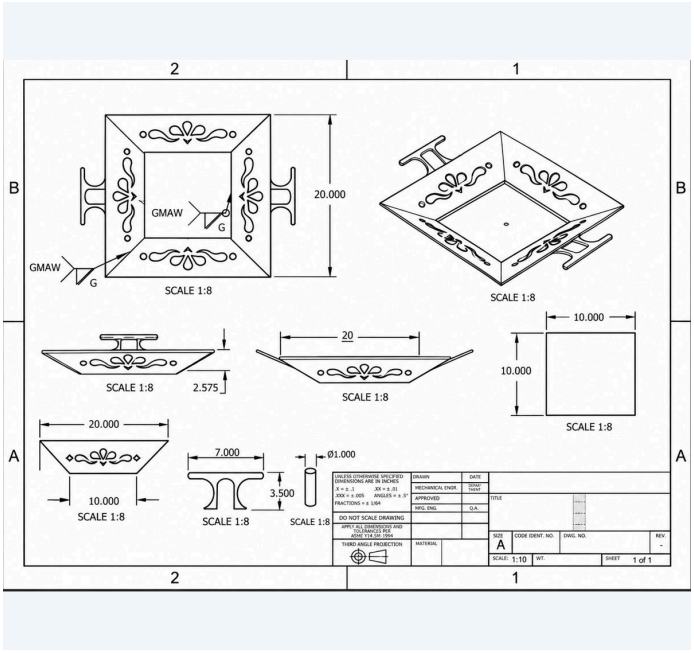
Current status

This is ongoing team work. No load-test results or competition performance are claimed at this stage.

Steel Fire Pit: Design to Fabrication

Independent design-and-build project completed through the Maxwell High School Welding Program, from CAD drawings to a finished steel fire pit.





Design and documentation

Developed the geometry in AutoCAD and Onshape, producing orthographic and isometric views with dimensions. The design includes a square center, outward-sloping panels, decorative cutouts, and integrated handles.

Fabrication and challenges

Worked with laser-cut steel components and GMAW welding. Aligning sloped panels and maintaining consistent joints required careful fit-up, sequencing, and repeated checks during assembly.

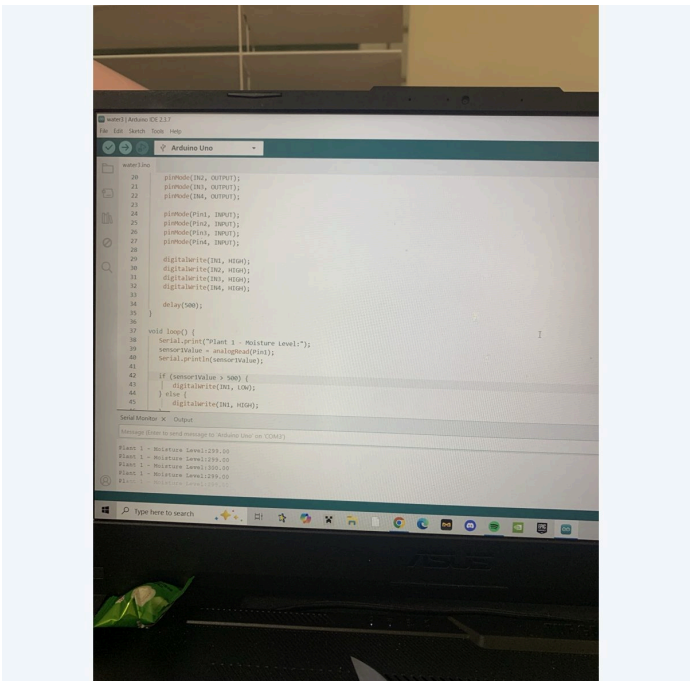
Outcome

Completed a finished fire pit and gained firsthand experience carrying a design through fabrication, assembly, and quality checks.

Arduino Smart Irrigation

Independently built an automated watering system for a snake plant using a soil-moisture sensor, Arduino Uno, relay, and water pump.





System logic

The soil-moisture sensor provides a reading to the Arduino. When the programmed condition indicates insufficient moisture, the controller activates the relay and pump to deliver water.

Calibration

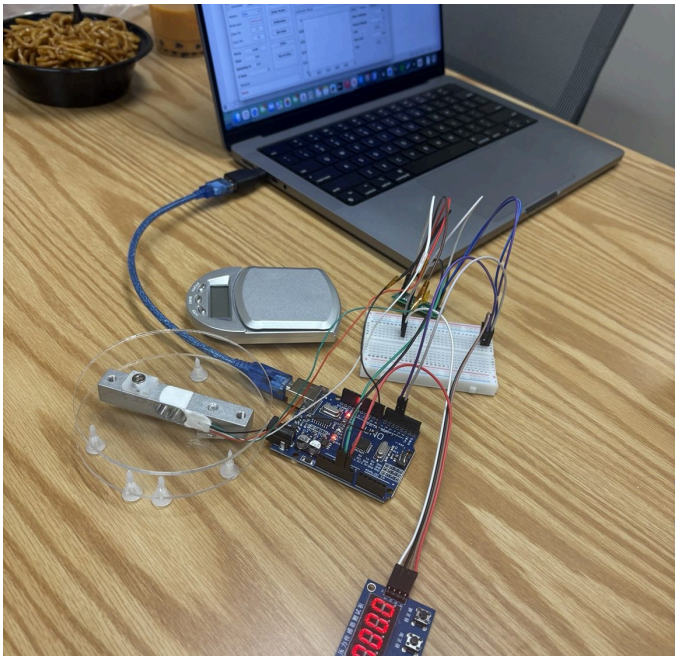
Set a sensor threshold of 500 for the prototype, reflecting the snake plant’s relatively low watering requirements. This threshold is specific to the sensor setup and requires calibration for different soils or plants.

Engineering skills

Integrated hardware, programmed conditional control, tested sensor behavior, and adjusted the system based on plant needs.

Arduino + MATLAB Calorie Scale

Team-developed food-weighing prototype connecting a load-cell sensor and Arduino to MATLAB for portion-based nutrition calculations.



```matlab
T = readtable('Data.txt');
avgVar2 = mean(T.Var2);
fprintf('%.1f g', avgVar2)
myInput = input("What type of food are you weighing?\na.Chicken Breast\nb.Ground Beef\nc.Porkbelly\n",'s');
if strcmpi(myInput,'a')
    fprintf("Chicken Breast\n")
    calories = avgVar2 * 165/100;
    protein = avgVar2 * 31/100;
    fat = avgVar2 * 3.6/100;
    carbs = 0;
    fprintf('For %.1f g of chicken breast,\n', avgVar2);
    fprintf('  Calories: %.2f kcal\n  Protein:  %.2f g\n  Fat:      %.2f g\n  Carbs:    %.2f g\n', calories, protein, fat, carbs);
elseif strcmpi(myInput,'b')
    fprintf("Ground Beef\n")
    fprintf("What is the lean/fat ratio?\n")
    leanPercent = input("Lean%?\n");
    fatPercent = input("Fat%?\n");
    if leanPercent + fatPercent == 100;
    calories = avgVar2 * leanPercent/100 * 131/100 + 9 * avgVar2 * fatPercent/100;
    protein = avgVar2 * leanPercent/100 * 25/100 + 0 * avgVar2 * fatPercent/100;
    fat = avgVar2 * fatPercent/100;
    carbs = 0;
    fprintf('For %.1f g of ground beef, (Lean: %.1f%%, Fat: %.1f%%):\n', avgVar2, leanPercent, fatPercent);
    fprintf('  Calories: %.2f kcal\n  Protein:  %.2f g\n  Fat:      %.2f g\n  Carbs:    %.2f g\n', calories, protein, fat, carbs);
    else
        error("Invalid input. Percentages must add up to 100%");
    end
elseif strcmpi(myInput,'c')
    fprintf("Porkbelly\n")
    calories = avgVar2 * 520/100;
    protein = avgVar2 * 10/100;
    fat = avgVar2 * 51/100;
    carbs = 0;
    fprintf('For %.1f g of Porkbelly,\n', avgVar2);
    fprintf('  Calories: %.2f kcal\n  Protein:  %.2f g\n  Fat:      %.2f g\n  Carbs:    %.2f g\n', calories, protein, fat, carbs);
else
    error('Invalid input. Please enter a, b, or c.');
end
```

My contribution

Set up the Arduino hardware, worked on load-cell calibration and raw-data transfer to MATLAB, and participated in integration and system testing.

How it works

A load cell and HX711 interface capture mass measurements. MATLAB processes the readings and scales nutrition reference values to the measured food portion.

Challenges and learning

Stable, repeatable readings required attention to wiring, leveling, calibration, and vibration. The project combined physical sensors with computational processing and teamwork.

Let’s connect.

Open to civil engineering internship and co-op opportunities for Summer 2027, particularly in geotechnical engineering, construction, and infrastructure.

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